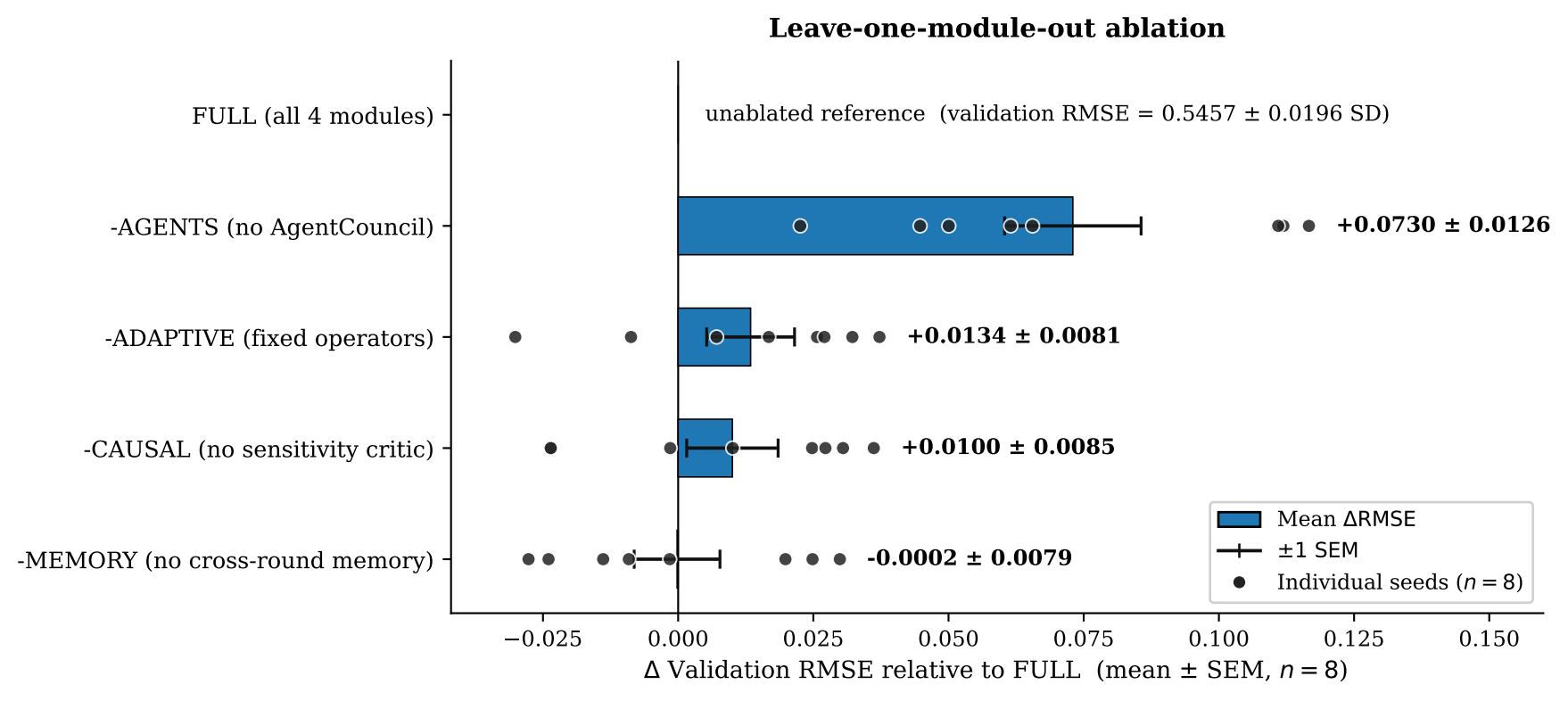
Leave-one-module-out ablation
FULL (all 4 modules)
unablated reference (validation RMSE = 0.5457 ± 0.0196 SD)
-AGENTS (no AgentCouncil)
+0.0730 ± 0.0126
-ADAPTIVE (fixed operators)
+0.0134 ± 0.0081
-CAUSAL (no sensitivity critic)
+0.0100 ± 0.0085
-MEMORY (no cross-round memory)
-0.0002 ± 0.0079
Mean ΔRMSE
±1 SEM
Individual seeds (n = 8)
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
Δ Validation RMSE relative to FULL (mean ± SEM, n = 8)